

MUHAMMAD HUZAIFA AFTAB

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EDUCATION

B.Sc. Mechanical Engineering | University of Engineering & Technology (UET) Taxila, Pakistan

Aug 2023 – July 2027 (Expected)

Relevant Coursework: Machine Design, CAD/CAE, Manufacturing Processes, Mechanics of Materials, Materials Engineering, Measurement & Instrumentation, Control Engineering, Thermodynamics, Fluid Mechanics, Numerical Analysis, Programming Fundamentals

RESEARCH EXPERIENCE

Final Year Research Project | Group Leader

EDM Machining of Aerospace Alloys Using Green Dielectric Fluids (HEC-NRPU Funded Project), UET Taxila | 2026 – Present

- Lead a four-member undergraduate research team investigating electrical discharge machining (EDM) of aerospace alloys using environmentally friendly green dielectric fluids, under faculty supervision, as part of a Higher Education Commission (HEC) National Research Program for Universities (NRPU)-funded initiative.
- Coordinate experimental planning, literature review, and technical documentation as group leader, applying structured research methodology to an advanced manufacturing and sustainable machining problem.

Research Intern | Group Leader, Manuscript Team

MultiOmics —Remote | 2026 – Present

- Lead a review manuscript team on “AI Applications in Additive Manufacturing: Opportunities, Challenges, and Future Trends,” coordinating research tasks and producing production-quality technical documentation ahead of journal submission.
- Conduct systematic literature review, critical analysis of peer-reviewed sources, citation verification, and technical writing to support evidence-based engineering argumentation.

ACEDAMIC PROJECTS

Programming & Automation | SOLIDWORKS API/VBA, Python, MATLAB

Personal & Academic Projects

- Built parametric SOLIDWORKS VBA macros for automated part/drawing generation (shafts); completed CAD Sharp LLC's VBA & API Programming Course.
- Certified in PDM API Write Add-ins (C#/VBA); built personal portfolio site, contributed to a Python fee-verification web app, and used MATLAB for engineering data analysis.

Automated Water Level Control System (PID Controller)

Complex Engineering Problem, UET Taxila | Arduino, MATLAB/Simulink, PID Control

- Developed an Arduino-based PID water-level controller with ultrasonic sensing and automated pump control, targeting 8 cm with a 3–8 cm operating range.
- Tuned $K_p = 2$, $K_i = 0.05$, $K_d = 0.01$ and achieved zero overshoot with ~ 236 s settling time in MATLAB/Simulink.

Cam-Driven Color Sorting Mechanism

Mechanism Design & Kinematic Analysis, UET Taxila

- Designed and tested an Arduino-based cam-follower system for automatic red/green color sorting, using SHM cam profiles and 2 independent 1-DOF mechanisms.
- Achieved 40 mm follower stroke, 125.7 mm/s maximum velocity, 789.6 mm/s² acceleration, and 95% sorting accuracy, with a safe 18.4° pressure angle.

Centrifugal Pump — Effect of Impeller Inlet Vane Angle on Performance

Fluid Machinery | Analytical & Mathematical Modeling

- Investigated the effect of impeller inlet vane angle (16°–30°) on centrifugal pump performance using a simplified analytical model, calculating outlet whirl velocity, impeller power, and pump efficiencies at each angle.
- Found that increasing the angle reduced outlet whirl velocity (18.90 → 12.59 m/s) and impeller power (95 → 63.2 kW), while increasing manometric efficiency (61.8% → 93.0%).

Electrically Actuated Spring-Loaded Centrifugal Governor

Machine Design-II | SOLIDWORKS & COMSOL Multiphysics

- Designed and analyzed a 500–3000 RPM centrifugal governor using SOLIDWORKS, combining mechanical speed regulation with electrical actuation and a spring-damper system.
- Performed theoretical and COMSOL analysis, obtaining 5,919 N centrifugal force per flyball, 0.033–0.038 m sleeve displacement, and 75.4 MPa arm stress, confirming stable motion and safe operation

Battery Pack Heat Transfer Analysis

ANSYS Fluent | Heat & Mass Transfer

- Developed a CFD model of 4 prismatic lithium-ion cells with water cooling, using 250 W/m³ heat generation and 0.05 m/s inlet velocity.
- Achieved 6–8% lower maximum temperature ($\sim 53 \rightarrow 49$ –50°C) by increasing cell spacing, with <2% mesh sensitivity and 10–15% correlation deviation.

WORK EXPERIENCE

Engineering Operations Intern

Wah Nobel | Aug 2026 – Present | *Multinational joint venture between Swedish and Saudi Arabian partners*

- Gaining hands-on exposure to chemical plant operations, manufacturing processes, industrial equipment, and plant utilities, including pumps, valves, and piping systems.
- Developing practical understanding of mechanical systems, industrial maintenance practices, process equipment, and safety procedures within a chemical manufacturing environment; collaborating with technical staff, observing engineering operations, and preparing technical documentation of plant activities.

Product Design Intern

Machadav | Jul 2026 – Aug 2026

- Completed a 4-week hands-on internship focused on product development, PCB enclosure design, CAD modeling, and 3D-printing prototyping; designed and refined PCB enclosures and mechanical components in CAD, optimizing for functionality, manufacturability, and 3D printing.
- Evaluated and developed mechanical mechanism concepts for product improvement, translating requirements into practical CAD designs.

Manufacturing Operations Intern

Heavy Mechanical Complex (HMC), Taxila | Jul 2025 – Aug 2025

- Completed a six-week rotational internship at HMC, Pakistan's "Mother Engineering Industry," gaining hands-on exposure to production planning, quality inspection, foundry, forging, CNC machining, and assembly.
- Developed practical understanding of large-scale manufacturing, precision engineering, and industrial production workflows, connecting mechanical design concepts with real-world manufacturing processes.

CAD/CAE Design Intern

National Aerospace Science & Technology Park (NASTP) | Jun 2025 – July 2025

- Modeled aerospace components and assemblies in SOLIDWORKS, maintaining tight dimensional accuracy for downstream manufacturability.
- Performed structural/FEA simulations in ANSYS to validate design integrity under load, supporting design optimization decisions.

TECHNICAL SKILLS

- **CAD & PDM:** SOLIDWORKS (CSWE), API/VBA Macros, PDM Add-ins, Solid Edge (CME), GD&T, Reverse Engineering, Assembly Design
- **Simulation:** ANSYS (FEA, Fluent/CFD), Structural & Thermal Analysis, Mesh Convergence, Nonlinear Analysis, HAP (HVAC)
- **Modeling & Analysis:** Fluid Machinery Performance, Parametric Calculations, Data Analysis & Visualization
- **Programming:** Python, MATLAB, VBA, Arduino, PID Control
- **Documentation:** Technical Writing, Literature Review, Engineering Reports

LEADERSHIP & COMMUNITY

Founder & Lead Mentor

MechMate Hub | Aug 2024 – Present

- Founded a 600+ member global CAD learning community; mentored 50+ students through SOLIDWORKS certification with successful outcomes.

Chairperson & Lead Mentor

SOLIDWORKS Club, UET Taxila | Jul 2024 – Present

- Founded and lead a 40+ member, 5-team volunteer club; host industry workshops and mentor members in CAD/design workflows.

Campus Ambassador

ASPIRE Institute & Siemens | Aug 2026 – Present

- Represent ASPIRE Institute as Campus Leader and Siemens AG Campus Ambassador, connecting students with global fellowship and engineering-tech opportunities.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Certified SOLIDWORKS Expert (CSWE) with 20+ SOLIDWORKS Certifications — Dassault Systèmes
- Completed MATLAB Onramp and Simulink Onramp — MathWorks
- Geometric Dimensioning & Tolerancing (GD&T) for Engineering — Udemy
- Certified Istec Vibration Training Category 1+ — ISTE International
- Google Project Management Professional Certificate V2 — Google / Coursera

Conferences Attended

- 3rd Global Science, Technology and Management Conference (GSTMC-25), University of Jhang — Jan 2025
- 1st International Conference on Advances in Mechanical, Material, Mechatronics and Energy Engineering (ICAME-24)
- 2nd UW-POFs Joint Conference on Academia's Collaboration with Industrial Organizations (CACIO-2022) — Dec 2022